THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 1: **Exploring the Investigative World of Science** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. An observation that cannot be repeated by others is: 1

(a) reliable **(b)** proven **(c)** a law **(d)** unreliable

2. Which question needs measurement to be answered? 1

(a) Do you like ice?
(b) Is ice pretty?
(c) How long does ice take to melt in a warm room?
(d) Is ice cold feeling good?

3. Which graph is best to show how temperature changes with time? 1

(a) line graph **(b)** pie chart only **(c)** list of names **(d)** no graph

4. Science differs from belief because it relies on: 1

(a) evidence from tests **(b)** opinions
(c) stories only **(d)** age of the idea

5. A lab report includes the following in the order: 1

(a) aim, apparatus, procedure, observations, conclusion
(b) conclusion, aim, apparatus, observation
(c) observation, aim, conclusion, apparatus
(d) apparatus, conclusion, aim

6. Reading 25.4 cm is measured with a ruler marked in mm. The precision of the instrument is: 1

(a) 1 m **(b)** 0.1 mm **(c)** 1 mm **(d)** 1 cm

7. The mean of 3, 5, 7, 9 and 11 is: 1

(a) 7 **(b)** 5 **(c)** 35 **(d)** 9

8. If a thermometer reads 2 degrees C when kept in melting ice, it has: 1

(a) no error **(b)** zero error **(c)** time error **(d)** parallax error

9. An investigator changes both the amount of water and the amount of sunlight for plants. This is: 1

(a) a hypothesis **(b)** not a fair test **(c)** a conclusion **(d)** a fair test

10. Which is the best first step if two groups get different results? 1

(a) compare methods and repeat **(b)** select the better-looking result
(c) average blindly **(d)** stop the experiment

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why do scientists use a control in an experiment? 2

12. Why must a hypothesis be testable? Give an example of one that is not. 2

13. A pendulum takes 27 s for 15 swings. Find its time period. 2

14. Differentiate between measurement and observation. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain how observation, hypothesis, experiment and conclusion are linked. 3

16. List the points to be remembered while reading the scale of a measuring cylinder. 3

17. Why are labelled diagrams useful in a lab report? 3

OR

Differentiate between an inference and an observation with an example from a kitchen. 3

18. A new medicine is claimed to work by one person. List three checks needed before accepting the claim. 3

OR

Suggest three ways to reduce errors while using a stopwatch. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Differentiate between precision and accuracy with an example from measurement of a table's length. 5

OR

Explain the importance of standard units. What problems arise if each person uses their own unit? Give an example. 5

20. Explain, with reasons, how each of the following improves an investigation: fair test, repetition, tabulation, graph, honest reporting. 5

OR

Describe the steps for collecting, tabulating and graphing data in an investigation of 'time for sugar to dissolve in water at different temperatures'. 5

END OF QUESTION PAPER — Total: 40 Marks

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